

Gender Discrimination in Competitive Online Games – Perceptions, Rumination, and Moderation

Ronja Fuchs, Jakob Droste, Alexander Dockhorn

Abstract—Competitive online gaming remains male-dominated, and toxicity disproportionately targets non-male players. Under the veil of anonymity, players with marginalized gender identities encounter discrimination, hate speech, and unwanted sexual advances. Addressing these issues requires understanding how players perceive gender-based harassment. This work investigates gender discrimination in Competitive Online Multiplayer Games through two surveys. In the first survey, 61 non-male players assessed gender-based discrimination and reported their use of moderation tools, including identity concealment or disclosure. Their responses yielded 171 statements describing forms of gender discrimination. In the second survey, 281 players across all gender identities rated these statements for severity and authenticity and reported how they use in-game moderation tools in response to discriminatory behavior. Our results indicate that gender discrimination is perceived as similarly severe across gender groups, but responses differ: players from marginalized groups report higher levels of rumination. Higher perceived severity predicts increased moderation tool use; male players show a stronger rise in muting offenders, whereas female players show a stronger rise in reporting them. Among previously targeted players, perceived severity does not predict gender concealment. These findings offer guidance for game developers seeking to design more inclusive gaming environments.

Index Terms—Player Experience, Online Harassment, Gaming Communities, Survey Studies

I. INTRODUCTION

Competitive Online Multiplayer Games (COMGs) are among the most popular games, and despite traditionally being considered a male-dominated space, female players' numbers are rising [1]. Riot Games reported that up to 40% of VALORANT players identify as female, compared to up to 30% in the League of Legends (LoL) player base [2]. Overall, around 34% of first-person shooter (FPS) players in the United States were reported to be female in 2024 [3]. Based on this, it is reasonable to assume that approximately one-third of the player base in these genres identifies as women in 2025. Gaming has been found to foster social interaction among players [1], [4]. However, female players alongside other marginalized groups frequently report experiences of harassment [5]. For women in particular, gender-based discrimination remains a common experience in competitive online environments. Despite continued advances and a growing number of initiatives aimed at reducing toxicity in games, the problem persists. Since toxicity against minorities heavily impacts their mental

health, previous work has underlined the importance of spaces free from prejudice, in which minority groups can meet and socialize [6]. If handled appropriately, COMGs have the potential to offer a wide range of mental health benefits to players, especially to those in minority groups [6]. Social support in particular is not only an important factor for mental health after experiencing harassment, but also an existential motivating factor for players to play games [5].

Although overall gamer demographics are relatively balanced, prior research suggests that female players remain a perceived minority in Competitive Online Multiplayer Games. Their perceived membership in a minority group often leads other players to assign them lower social status by default [4], [5], [7]. The growing number of women in the traditional male-dominated space is not always welcomed by male users, and hence sparks a problem of sexual harassment in the gaming community [6]–[9]. Further than that, female players are constantly reminded of their perceived intrusion of the intended male subject position within online gaming spaces through sexual harassment and gender discrimination [7]. Spaces dedicated to geek culture, which include gaming communities, may exhibit the tendency to view women as either objects of sexual desire or unwelcome intruders [8].

To overcome gender discrimination in COMGs, we must understand how it is perceived by the discriminated minority as well as the overall community, regardless of gender identity. Prior research has been conducted to observe the overall problem of harassment of women in gaming [7], [9]–[12] as well as the normalization of toxicity in gaming spaces [6], [13]. To build upon this research, we specifically examine how players of different gender identities are influenced by gender discrimination against women, to what extent it makes them alter their behavior online, and in what way it discourages or even hinders them from playing. We therefore examine how players of all genders perceive the severity of gender discrimination toward women and how these perceptions relate to behavioral changes in response. Furthermore, we aim to understand how the perceived severity impacts rumination and consequent withdrawal from games. In this context, we conducted two studies with 61 and 281 participants, respectively. In both studies, participants were invited to share their perceptions of gender discrimination and harassment directed toward women.

This paper is an extended version of a paper originally published in the *2025 IEEE Conference on Games*, in which we analyzed how players of all other genders perceive gender discrimination in COMGs [14]. We primarily analyzed players' general perception of gender discrimination in COMGs and how

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severe they perceive gender discrimination categories and real examples to be. Furthermore, we examined which players tend to ruminate, i.e., dwell on negative feelings after witnessing or experiencing gender discrimination. In this extension, we report on additional survey data, which were not discussed in the original paper. In particular, we observe and discuss how gender discrimination and players' perception of it influence players' usage of in-game moderation tools as well as choices to conceal or reveal their gender.

Throughout this paper, references to "female players", "male players", or any other gender designation refer to gender identity as self-identified by the individuals, not to biological sex or sex assigned at birth.

This paper proceeds as follows: Section II reviews literature on gender discrimination and harassment in online gaming; Section III outlines the research questions and methodology; Section IV presents our findings; Section V discusses implications and validity threats; and Section VI concludes with future research directions.

II. BACKGROUND AND RELATED WORK

A. Harassment in Online Games

The reasons behind the harassment of marginalized groups in COMGs have been widely discussed in prior research [6], [7], [9], [12], [13], [15]–[17]. Fox and Tang [7] found that social dominance orientation, the desire for power over women, and the need for heterosexual self-preservation are significant predictors of video game sexism. The authors explain that especially masculine behavior, such as competitiveness or aggression, is often rewarded in gaming environments, providing male players with opportunities to assert social dominance that may not be feasible in the physical world. They emphasize, that even if gaming as an activity continuously gains attraction among women, as sexism and its expressions become increasingly prevalent, women may be pushed into silent participation or driven away entirely.

Prior research offers numerous frameworks for explaining the pervasiveness of discrimination and harassment in online spaces. The online disinhibition effect described by Suler [15] suggests that inherent characteristics, including anonymity, minimized authority or lack thereof, and a hypermasculine atmosphere, may facilitate harassment online. Furthermore, the Social Identity Model of Deindividuation Effects (SIDE), as proposed by Postmes et al. [18], argues that the anonymity of online interactions leads users to downplay their individual identities and instead conform to salient group norms, reinforcing collective behaviors. Within the veil of anonymity, people tend to feel less vulnerable and choose to act out more, especially when provided the opportunity to separate their online actions from their real-life character, as described by Suler [15]. Male players may consequently engage in toxic and discriminatory behavior as a way of conforming to perceived contextual standards and affirming group cohesion. Fox and Tang [9] add that this includes the realm of COMGs, and explain that male players in the male-dominated space of gaming are likely to defer to their masculine identity rather their individual self. Players downplay their individual identity in favor of

defending the prevailing social identity (masculine, competitive, and aggressive) [9]. Even if they might not act the same way in the real world, players often practice moral disengagement by employing cognitive operations that allow them to justify, rationalize, and downplay problematic behavior [19].

Examples of this effect can be found in many popular COMGs. According to Ma et al. [6], FPS games may be the most toxic environment for women. They suggest SIDE to be a main cause. This is supported by Maharani et al. [16], who describe how gender discrimination is a universal experience for women in VALORANT. Yet online gaming is not exclusively harmful. Ma et al. [6] offer a dual perspective: while prejudice and discrimination are normalized in some gaming spaces, online games and communities may serve as healing environments. Players of minorities may find safe spaces and a wide range of mental-health benefits in those communities.

Toxicity extends beyond specific genres: around 70% of *League of Legends* matches contain toxic behavior [17], while 80–85% of players across games have witnessed or experienced it [20], [21].

However, as toxicity is highly subjective, players may hold different opinions on what constitutes toxic behavior. The same behaviors that are perceived as inappropriate or even harmful by some, may be interpreted as tolerable or even enjoyable by others [13]. Souza et al. [12] conducted a study to find out how players of different genders perceive and experience toxic behavior in online games. Regardless of gender, their participants unanimously identified toxic behavior as prevalent in games, although male players most reported verbal aggression while female players predominantly experienced skill underestimation and sexual harassment. Beyond the general persistence of toxicity in gaming, women receive disproportionately more negative comments than their male counterparts. Kuznekoff and Rose [22] corroborate this, finding that women receive three times as many negative comments as men in voice chat, with players reacting measurably more negatively to a female voice than a male one. While harassment affects players of all genders, women and girls disproportionately experience more severe forms of harassment. As Souza et al. [12] focus on the differences of toxicity as experienced by players of different gender identities, we aim to explore the discrimination of women in gaming, more specifically in this work.

B. Effects of Harassment in Online Games

The impact of players being targeted by sexual harassment varies among players, leading them to adopt different coping mechanisms. According to Adinolf and Türkay [23], ignoring or muting other players is the most common coping strategy, which inadvertently allows toxic behavior to persist unchecked. Frommel and Mandryk [24] corroborate this, but note that changing in-game appearance or username was far less common than they expected, ranking as the second-to-least used strategy overall. However, Drenten et al. [25] argue that any such change leads to a feedback loop, in which marginalized groups use coping strategies that in turn solidify the perceived dominance of the majority group. Women, McLean and Griffiths [5]

note, commonly report feelings of anxiety and loneliness caused by the lack of societal support and the prevalence of harassment. This dynamic often leads them to play alone, remain anonymous, or frequently change groups. It may further contribute to the isolation of marginalized players, reduce their visibility in gaming spaces, and reinforce the perception that competitive gaming remains a male-dominated environment. As a result, toxic behaviors may go unchallenged, while the absence of diverse voices in gaming communities limits efforts to foster inclusivity. Given that emotional responses to harassment in gaming have been extensively documented in previous research, our study focuses on the players' own perception of harassment, its severity, and its relation to their self-concept.

Features like player moderation tools, with which players may mute, report, or block other players, are widely adopted in COMGs. Adinolf and Türkay [23] found that reporting tools are underused due to limited transparency and perceived ineffectiveness, whereas Kowert and Cook [20] identify reporting as the most common response to toxicity. Türkay et al. [26] found that players focus mainly on the disruption that toxicity brings to the play, rather than its impact on the victims, with coping strategies primarily revolving around preserving the gameplay rather than eliminating toxicity. They add that forms of toxicity that discriminate against marginalized players were not recognized by people who were not part of the targeted group.

Harassment in online games, as in real life, can cause players to dwell on negative emotions and anxiety long after the actual incident [9]. Fox and Tang [9] explain that the frequency of sexual harassment in online video games contributes to this rumination. The authors further state that rumination over harassment can lead to withdrawal from the game and, in some cases, from gaming altogether. This dynamic results in the harassment of those perceived as outsiders, particularly women, aligning with the SIDE [18] model. Additionally, women respond to general harassment differently than they do to sexual harassment [9]. This suggests an important distinction between being harassed as part of a majority group and being targeted as a member of a minority or otherwise stigmatized group. As a result, Cote [11] identifies five coping strategies used by female players, with leaving online gaming being the most common. Other strategies were avoiding strangers, camouflaging gender, deploying skill and experience, and personality strategies such as the adoption of aggressive behavior. Even beyond casual play, the consequences extend to professional esports. Crothers et al. [27] highlight that female players in competitive esports face systemic gender discrimination, marked by exclusion, stereotyped roles, and the normalization of sexual objectification.

Since the study conducted by Fox and Tang [9], the number of female gamers has continued to rise. According to The Entertainment Software Association (ESA) [28], women accounted for 37% of overall gamers in the USA in 2017, a figure that increased to 48% by 2025 [29]. Given this shift, we aim to explore how these effects apply to players today and whether they persist despite the increasing number of female gamers.

III. METHODOLOGY

A. Research Goal and Questions

This paper aims to evaluate how players of different genders perceive gender discrimination against female players in online games. Additionally, we seek to comprehend how the perceived severity of this gender discrimination affects players' user experience.

To reach our research objective using suitable statistical measures, we apply the Goal-Question-Metric (GQM) method as outlined by Wohlin et al. [30], and formulate the following research questions:

- RQ1** How do players perceive gender discrimination against female players in Competitive Online Multiplayer Games, and how do these perceptions differ across gender groups?
- RQ2** How does the perceived severity of gender discrimination among players relate to rumination, and how does this relationship vary across gender groups?
- RQ3** What is the impact of gender discrimination-related rumination on player experience in Competitive Online Multiplayer Games, and how does rumination relate to discouragement from playing Competitive Online Multiplayer Games and consequent withdrawal across gender groups?
- RQ4** How does the perceived severity of gender discrimination among players relate to their usage of moderation tools and how does this relation differ between gender groups?
- RQ5** How does the perceived severity of gender discrimination relate to discriminated players altering their behavior or changing their information in COMGs to hide their gender?

To address these questions, we used five sets of metrics, which were evaluated using online questionnaires. The metrics can be found in our supplementary material [31].

B. Survey Design

To address the RQs, we carried out two consecutive surveys: the first gathered examples of gender discrimination from female players, and the second had players of all genders assess these examples. The questionnaires can be found in our supplementary material [31]. An overview of the surveys and their content is displayed in Figure 1.

1) *Common Methods in both Surveys:* Both surveys included a demographic questionnaire. Participants were asked which COMGs they played regularly and the number of hours they spent playing each week. Additionally, they were asked to specify whether they were actively involved in gaming communities, participated only passively, or were not involved at all. We also inquired if they fully identified as a "gamer", identified as one only in certain contexts, or did not consider themselves a "gamer" at all. Participants were further asked to indicate if they had ever left a gaming group or community due to gender discrimination. In the second part of each survey,

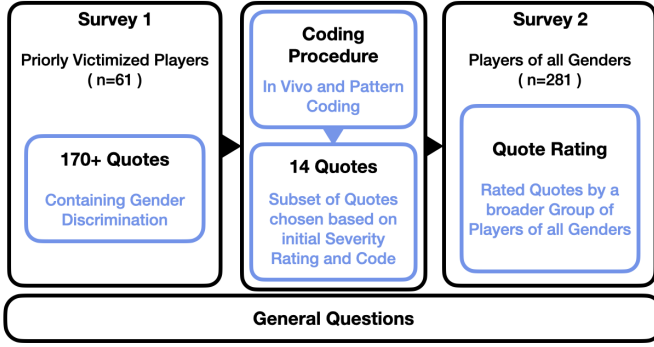


Fig. 1: Overview of Surveys and Coding Procedure

participants were asked to rate the severity of various types of gender discrimination on a 7-point scale. These types were derived from prior research [5], [7], [11], [16], [22], [32], [33].

The severity rating assessed the personal impact of these categories on the participants and the extent to which they believed these issues violate the Terms of Service (ToS) of the games they frequently play. Participants responded to statements regarding their sense of belonging in the overall gaming community and the state of gender discrimination within the community, using 5-point Likert scales. They were also asked about their use of in-game player moderation tools, such as reporting or muting a player. Additionally, participants indicated their level of agreement (on a 5-point Likert scale) with exhibiting specific behaviors after encountering sexist statements, such as defending the victim or ignoring the comment. Finally, all participants were asked to provide their gender identity, age range, and the locations of the servers they typically play on.

2) *Survey 1: Methodology and Participant Selection:* In our first survey (S1), only women were invited to complete an online questionnaire. Participants were recruited through university courses, the authors' personal networks, women-only Discord communities, and collegiate esports communities. To gain a comprehensive understanding of the participants' experiences, we asked them to share quotes involving gender discrimination that were directed at them or others while they were present (see Figure 1). Participants were then asked to assess the severity of the impact of these quotes on themselves or in terms of their violation of the game's Terms of Service (ToS).

To explore how players interact with the broader community, participants were further asked whether and how they adjust their behavior during gameplay in response to gender discrimination, such as by changing their username or altering their voice.

3) *Survey 2: Methodology and Participant Selection:* In the second survey (S2), players of all genders were invited to participate. To achieve this, the survey was shared in gaming community Discords, university courses, the authors' personal networks, and collegiate esports communities. Additionally, it was posted in gaming subreddits for popular COMGs that

permitted the survey, such as r/VALORANT and r/cs2).¹

The quotes of players that were provided by participants of S1 were utilized in S2 (see Figure 1). Since over 170 submissions made full inclusion impractical, we randomly selected two quotes per severity rating as assigned by S1 participants, resulting in a total of 14 quotes (cf. Table III). After completing S1, the quotes were categorized, and for each of the three most prevalent categories, we ensured that at least one additional quote was included. All participants in S2 were asked to rate the severity of the combined 14 quotes. Although all quotes were taken directly from S1, we inquired whether they thought each quote was genuine or fabricated by us for the study, allowing us to assess whether players' expectations of harassment align with its actual prevalence.

C. Coding Procedure

a) *Analysis of Participants' Statements:* All examples of gender discrimination collected were analyzed through two coding cycles, using *in vivo* coding for the first cycle and pattern coding for the second. The entire coding process followed the guidelines set by Saldaña [34]. Initially, *in vivo* coding was performed by one researcher, who removed survey responses that lacked any form of gender discrimination and condensed the remaining responses to the relevant content. In the second coding cycle, three researchers were involved. Firstly, one researcher reviewed the initial coding results and developed a pattern coding scheme, which was then evaluated and discussed with a second researcher. After the codes were refined, the final pattern coding scheme was applied by three researchers to classify the statements into categories of gender discrimination. Importantly, our coding procedure followed a multi-label approach, allowing each statement to include multiple categories of gender discrimination.

b) *Inter-rater Agreement:* To assess inter-rater agreement for the pattern coding, we calculated both Fleiss' Kappa [35] and the proportional agreement among the coders. Proportional agreement refers to the ratio of statements on which the raters agreed to the total number of statements. The proportional agreement was found to be 76.2%, while Fleiss' Kappa was 80.6%. According to Landis and Koch [36], this indicates a substantial to almost perfect level of agreement.

D. Data Availability Statement

To ensure that our findings can be verified and reproduced, we offer supplementary material [31]. This includes the survey questionnaires, survey results, and details of our coding procedure. To protect participant privacy, all published data has been anonymized.

¹To meet requirements of the subreddit r/VALORANT, a survey version was adapted to focus solely on VALORANT, excluding other COMGs and non-existent player moderation tools. Since the broader study included VALORANT players, their responses were merged. Additional questions on VALORANT-related behavior were included but are only reported in the supplementary materials, as they do not directly relate to our metrics.

TABLE I: Age, Primary Region, and Gender Identity of Participants

Category	S1(%)	S2(%)	Category	S1(%)	S2(%)
Age and Primary Region			Gender Identity		
Age (18–25)	70	54	Female	91	28
European Server	94	69	Male	–	61
NA Server	6	25	Agender	4	1.5
Other Regions	<1	10	Genderfluid	2	1

TABLE II: Regularly Played Games and Gaming Habits of Participants

Category	S1(%)	S2(%)	Category	S1(%)	S2(%)
Regular Games			Gaming Habits		
LoL	56	27	>5 h/week	90	86
VALORANT	35	63	>20 h/week	29	27
Counter-Strike 2	9	11	Prefer PC	98	92
Other Games	–	61			

TABLE III: Combined Severity Ratings: Participants’ Ratings and Gender-Specific Ratings vs. Original Severity

ID	Quote Text	Overall	W	M	NC	S1
Q1	“are you female”	2.87	3.16	2.68	3.32	1
Q2	“Boosted Egirl”	4.09	4.30	3.98	4.21	2
Q3	“Do you have Instagram? Snapchat?”	3.70	3.79	3.68	3.78	3
Q4	“I didn’t know we had a woman on our team. Come here! (proceeded to flirt)”	4.75	4.67	4.79	4.79	3
Q5	“slut, whore, bitch”	5.59	5.64	5.61	5.43	4
Q6	“u have big boobas”	4.24	4.23	4.34	3.79	4
Q7	“Go back to the kitchen”	4.93	4.94	4.90	5.07	4
Q8	“Hey [in-game character], are you part of the alphabet community?”	3.72	4.02	3.60	3.57	5
Q9	“Kissing sounds”	4.28	4.42	4.28	3.91	5
Q10	“I was busy thinking about getting pegged. let’s make out right now”	5.23	5.05	5.36	4.89	6
Q11	“Give me your OnlyFans!”	4.81	4.83	4.88	4.43	6
Q12	“You play like a girl”	3.97	4.01	3.97	3.89	6
Q13	“your voice is makes me hard, can you suck on my dick and have e sex with me”	6.25	6.27	6.32	5.86	7
Q14	“I hope you get raped, killed and pissed on your dead body”	6.76	6.85	6.78	6.46	7

IV. RESULTS

Participants of both surveys were notified that their results would only be analyzed in case they submitted their answers at the end of the survey. Hence, answers from unfinished surveys were not included. All responses to matrix-style questions were analyzed for pattern-based answering, such as straight- and pattern-lining. The exclusion threshold was determined through an outlier analysis of patterned responses for each survey.

A. Demographics

Overall, 342 participants filled out the surveys (S1: 61, S2: 281). 16 submissions (S1: 6, S2: 10) were excluded due to presenting pattern responses, and thus 326 submissions were analyzed. The most common age, primary region, and gender identity of participants are displayed in Table I. Regularly played games and gaming habits are displayed in Table II.

B. Perception of Severity of Gender Discrimination

Participants rated the severity of eight types of gender discrimination.² S1 participants averaged 5.38, while S2 participants averaged 4.93. In S2, male participants rated higher (5.02) than female participants (4.87) and gender non-conforming participants (4.57). A Wilcoxon-Mann-Whitney test ($p > 0.09$) found no significant difference between gender groups in rating specific types of gender discrimination ($\alpha = 0.05$). Groups were defined by gender and type of

discrimination. S1 participants reported experiencing various types of gender discrimination in the past month, with frequency weakly correlating with severity ratings ($r = 0.35$). Figure 2 shows the average severity and percentage of occurrence of types of gender discrimination by gender identity.

In S1, 42 of 61 participants submitted a total of 171 quotes from past gaming experiences. Selected quotes with participant-assigned severity ratings are shown in Table III. Quotes were categorized into eleven pattern codes, with eight quotes remaining uncategorized, as detailed in Table IV.

Table III presents the survey quotes alongside their average perceived severity. Female participants rated quotes slightly higher (4.73) than male participants (4.65), while gender non-conforming participants gave the lowest average rating (4.53). A Wilcoxon-Mann-Whitney test ($\alpha = 0.05$) compared severity ratings across gender groups, defined by gender and specific quotes. Significant differences were found between female and male participants for Q1 and Q8.

In S2, male participants guessed 74% of examples were real, female participants guessed 73%, and gender non-conforming participants guessed 70%.

C. Rumination and Gender Discrimination

Participants were asked to state whether they ruminate after a game in which a player makes a comment that includes gender discrimination. 34% of participants agreed or strongly agreed to ruminate after such a game (S1: 70%; S2: 25%), while 47% of participants disagreed or strongly disagreed (S1: 11%; S2: 56%). In S2, male participants mostly felt neutral or disagreed, while female participants tended to feel neutral or agree to ruminate. Gender non-conforming participants generally agreed or remained neutral.

Rumination and perceived severity were moderately correlated in S1 ($r = [0.24, 0.54]$) but only weakly in S2 ($r =$

²Types are “Players Trolling or Throwing a Game because of someone’s Gender” (T1), “Players Leaving a Game because of someone’s Gender” (T2), “Verbal Harassment with Name-Calling and Insults” (T3), “Verbal Harassment with Inappropriate Sexual Messages” (T4), “Patronizing (Assuming Skill Level based on Gender) or Dismissive Comments” (T5), “Unprompted Flirting or Unwelcome Romantic Advances” (T6), “Doxxing or Targeted Harassment” (T7) and “Microaggressions” (T8).

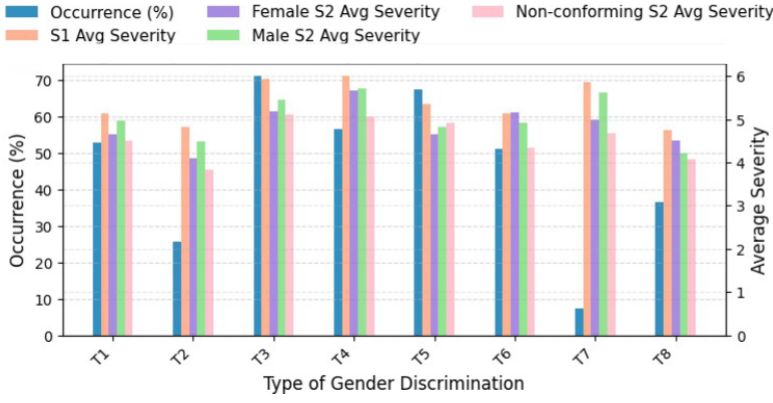


Fig. 2: Rated Average Severity and Occurrence of Types of Gender Discrimination

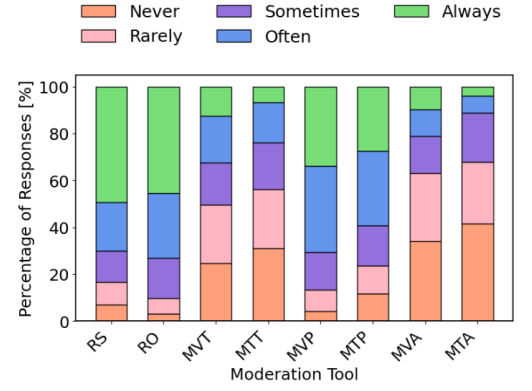


Fig. 3: Player Moderation Tool Usage after Gender Discrimination in S2

TABLE IV: Pattern Codes and their Occurrences.

Code	# Occurrences
Order to go into the kitchen or make a sandwich	43
Patronizing, "Game Over" mentality	43
Gender-specific insults	39
Sexual offers or advances	23
Messages with general sexual intent	18
Requesting to show the body, or asking for OnlyFans	10
Hate against LGBTQIA+	10
Asking for personal information	9
Sexual threats, rape threats	8
No Code	8
Derogatory sexualization	3
Gender-specific messages	1

TABLE V: Notable Pairwise Pearson Correlations Between Rumination and Perceived Severity for S1 and S2

Type	S1 (<i>r</i>)	S2 (<i>r</i>)
Verbal Harassment with Name-Calling and Insults	0.55	0.29
Verbal Harassment with Inappropriate Sexual Messages	0.54	0.28
Microaggressions	0.45	0.30
Players Leaving a Game Due to Someone's Gender	0.24	0.10
Doxing or Targeted Harassment	0.27	0.04

[0.04, 0.29]). Notable correlations with specific discrimination types are shown in Table V.

Overall, participants' severity ratings of all genders showed negligible to moderate correlations with rumination (W: $r = [0.09, 0.4]$; M: $r = [0.09, 0.26]$; NC: $r = [0.005, 0.534]$). The most notable of those are displayed in table VI.

To facilitate group-based comparisons between participants that gave higher than average ratings and those that gave lower than average ratings, we divided them into high-severity and low-severity groups. Levene's test indicated unequal variances (S1: $W = 9.30, p = 0.0031$; S2: $W = 37.36, p < 0.0001$), so a Welch's t-test was used, it showed a significant difference in rumination between the low- and high-severity group (S1: $t = 2.49, p = 0.0169$; S2: $t = 3.90, p = 0.0001$).

Rumination showed weak-to-negligible correlation with correctly identifying gender discrimination examples (see Table VI). For Q14, the most severely rated example, this correlation was negligible among male participants only.

TABLE VI: Notable Pairwise Pearson Correlations Between Rumination and Study Variables per Gender Identity

Variable	W (<i>r</i>)	M (<i>r</i>)	NC (<i>r</i>)
S2 Overall Severity Rating	0.20	0.28	0.43
S2 Severity: Name-Calling & Insults	0.41	0.16	0.53
S2 Severity: Players Leaving a Game	0.09	0.18	0.12
S2 Overall Moderation Tool Usage	0.36	0.26	0.23
S2 Muting all Players' Voice Chat	0.48	0.35	0.35
S2 Muting all Players' Text Chat	0.47	0.16	0.16
Quote Realness Identification (All Quotes)	0.10	0.04	0.20
Quote Realness Identification (Q14)	0.38	0.02	0.34

TABLE VII: Profile Adjustments Related to Gender Discrimination in S1

Prior Profile Adjustment and Voice Chat Usage (S1)	(%)
Username changed	45.5
Profile details modified (e.g., pronouns, descriptions)	32.7
Voice settings adjusted	16.4
No changes made at all	36.4
Significantly affected use of voice chat	29.1
Completely affected use of voice chat	47.3

D. Effects of Rumination and Gender Discrimination

Table VII presents profile adjustments made before the survey to conceal their gender identity, as well as affected voice chat usage in the S1 subgroup. Half of S1 participants (51%) reported leaving a gaming community due to gender discrimination, 23% in S2. Additionally, 59% of S1 participants agreed that gender discrimination discourages them from playing COMGs, 28% of S2 participants felt the same.

Levene's test confirmed equal variances between the low- and high-severity groups in S1 for player moderation tool usage; subsequent ANOVA revealed no significant differences between them ($F = [0.12, 0.72], p = [0.1, 0.72]$). No significant correlations were found between participants changing profile settings to conceal their gender, such as username changes, and the communication channels in which discrimination occurred, including team chat, all-player chat, or team voice chat. Levene's test confirmed equal variances between the low- and high-severity groups in S1 for the media through which participants experienced gender discrimination; subsequent

TABLE VIII: Significant Differences between Low- and High-Severity Groups regarding Moderation Tool Usage in S2

(Gender) Moderation Tool	Test	F or t	p
(all) Reporting when others were targeted	ANOVA	8.07	0.005
(all) Reporting when personally targeted	Welch's t-test	-2.36	0.02
(all) Muting Single Player's Voice Chat	Welch's t-test	-3.22	0.001
(all) Muting Single Player's Text Chat	Welch's t-test	-2.58	0.01
(W) Reporting when personally targeted	Welch's t-test	-2.38	0.02
(W) Reporting when others were targeted	Welch's t-test	2.16	0.04
(M) Muting Team Voice Chat	ANOVA	6.22	0.01
(M) Muting Single Player's Voice Chat	ANOVA	5.98	0.01
(M) Muting Single Player's Text Chat	ANOVA	4.78	0.03
(NC) Muting Single Player's Voice Chat	ANOVA	5.02	0.04

ANOVA revealed no significant differences between them.

Just over half of the participants felt a sense of belonging in the gaming community (S1: 54%; S2: 52%). An ordinal logistic regression found gender was not a statistically significant predictor of belonging (S1: $r = 1.034, p = 0.087$; S2: $r = 0.2581, p = 0.207$). While S1 showed some evidence of an effect, the null hypothesis could not be confidently rejected.

Most participants felt the gaming community was not welcoming to women (S1: 68%; S2: 57%). In S1, all participants experienced at least one of the given types of gender discrimination in the past month, with 42% reporting at least four out of eight types.

A moderate negative correlation existed between sense of belonging and altering identity settings to conceal gender ($r = -0.31$). Chi-square analysis showed a significant link between sense of belonging and changing usernames to hide gender ($\chi^2 = 11.35, p = 0.010$), but not with other methods like changing profile information.

Figure 3 shows how often S2 participants reported using certain player moderation tools after witnessing or experiencing gender discrimination.³ In both S1 and S2, participants used reporting mechanisms the most, while muting a single player's chat or voice was the second most used tool.

Participants in S1 were grouped into low- and high-severity raters (see Section IV-C). Levene's tests indicated homogeneity of variances for most moderation-tool measures, and ANOVA found no significant group differences in overall tool usage. The sole exception was reporting on behalf of others: Levene's test indicated unequal variances ($W = 8.73, p = 0.01$), and Welch's t-test showed a significant group difference ($t = 2.60, p = 0.015$). In S2, participants were similarly divided into low- and high-severity groups; all significant differences in moderation tool usage between these groups are reported in Table VIII, with Welch's t-tests applied where Levene's test indicated unequal variances and ANOVA otherwise.

Grouping all participants of S2 into high- and low-severity groups based on their rating of the examples revealed no significant differences regarding player moderation tool usage.

Pairwise correlation analysis revealed a weak to moderate association between rumination and player moderation tool

usage (S1: $r = 0.24$; S2: $r = 0.27$). In S2, the correlation was moderate among female participants ($r = 0.36$) but weaker among male ($r = 0.26$) and gender non-conforming ($r = 0.23$) participants. Moderate correlations were observed for muting all players in voice (W: $r = 0.48$; M: $r = 0.35$; NC: $r = 0.35$) and text chat (W: $r = 0.47$; M: $r = 0.16$; NC: $r = 0.16$). Overall, female participants exhibited stronger correlations between rumination and moderation tool usage, particularly for muting and reporting features (mean $r = 0.36$), compared to male ($r = 0.26$) and gender non-conforming ($r = 0.23$) participants. A moderate correlation was found between rumination and the feeling of discouragement from playing COMGs among participants (S1: $r = 0.47$; S2: $r = 0.56$). In S2, the correlation was strongest among gender non-conforming participants ($r = 0.76$), followed by female participants ($r = 0.62$), while male participants showed a notably weaker correlation ($r = 0.30$).

An ordered logistic regression found no strong evidence that participation level in communities outside the game influences rumination (S1: $r = 0.03, p = 0.916$; S2: $r = 0.11, p = 0.478$), including when analyzed by gender in S2 (W: $r = 0.03, p = 0.93$; M: $r = 0.195, p = 0.33$; NC: $r = -0.29, p = 0.48$).

V. DISCUSSION

A. RQ1: Perception of Gender Discrimination

The gender distribution in our study aligns with current statistics on female representation in COMGs, with approximately a third of participants identifying as women [3], [29], offering an up-to-date perspective, while previous research on harassment and rumination was conducted when the number of women was lower. In 2017, Fox and Tang [9] found that players of all genders acknowledged the harassment of women in games as a problem. Our findings indicate that this perception remains unchanged, contrasting Türkay et al. [26] who suggested that forms of toxicity that discriminate against marginalized players were not recognized by the people outside the targeted group. Male participants, despite constituting the majority group, rated the severity of gender discrimination as severe as female and gender non-conforming participants, both in their assessments of discrimination types and of the concrete examples provided in S1. Female participants rated the real examples marginally more severe than male participants (1% more severe on average). This might be a reflection of the fact that, having encountered such situations more frequently, concrete instances feel more severe to them than abstract categorical descriptions. While the general acknowledgement of toxicity as a problem aligns with both Fox and Tang [9] and Souza et al. [12], Fox and Tang further suggested that male and female players perceive harassment differently. Our findings offer partial support for this: although the frequency of experienced harassment has only a weak influence on overall perceived severity, it does shape how players assess the severity of specific harassment types. This is particularly relevant given that female players are disproportionately exposed to harassment [22], meaning that differences in type-specific severity perceptions are likely to have a more pronounced impact on their experience. This goes in line with Souza et al. [12], as they report that women experience different types of harassment than male players,

³The player moderation tools are reporting when personally targeted (RS), reporting when others were targeted (RO), muting voice chat of the team (MVT), muting text chat of the team (MTT), muting the offender's voice chat (MVP), muting the offender's text chat (MTP), muting everyone's voice chat (MVA), and muting everyone's text chat (MTA).

namely more sexual harassment and underestimation of their skills.

Although Türkay et al. [26] found that players' main focus when encountering toxicity was the impact it had on the game rather than its victims, our results indicate that players perceive the act of purposely losing through trolling or leaving a game as less severe than types of discrimination that primarily include direct communication with the target, at least when the reason is gender discrimination. This suggests a shared perception between players across all gender identities that harm carried through direct communication is seen as more consequential than indirect gameplay trolling.

Despite shifts in gender distribution, the premise of similarly perceived severity across genders remains consistent. The highly subjective nature of toxicity and harassment perceptions, as highlighted by Aguerri et al. [17], is less consequential in the context of gender discrimination specifically, as players across all genders converge in their assessments of its severity.

Finding 1: Players across all gender groups recognize gender discrimination against female players as a persistent issue. **Despite shifts in gender distribution, perceived severity of harassment remains consistent across genders**, with only slight differences based on exposure.

B. RQ2: Perceived Severity and Rumination

Generally, participants in the broader group reported little tendency to ruminate following experiences or observations of gender discrimination. Building on Fox and Tang [9], our findings however suggest that rumination on harassment experiences varies by gender. Female and gender non-conforming participants exhibited higher levels of rumination compared to male participants. This finding aligns with previous research indicating that marginalized groups respond to such experiences differently than the dominant group [5], [9], [11], [17].

Our analysis indicates that players who perceived gender discrimination as more severe exhibited higher levels of rumination, with this effect being more pronounced in S1 than in S2. Rumination was especially common among previously victimized participants, of whom 70% reported rumination following such experiences, compared to only 25% of the broader participant group. Notably, "Verbal Harassment with Name-Calling and Insults" and "Verbal Harassment with Inappropriate Sexual Messages" were rated as the most severe and showed some of the strongest correlations with rumination. This is consistent with broader evidence that verbal harassment remains one of the most prevalent forms of gender-based discrimination in COMGs, disproportionately affecting players from marginalized groups [6], [7], [9], [22], and is especially salient given that women receive substantially more negative comments than male players [22]. Together, these findings align with our hypothesis that more frequently encountered forms of harassment tend to be perceived as more severe, while also supporting Aguerri et al.'s [17] emphasis on the subjective nature of toxicity in gaming. More broadly, they point to the existence of underlying factors that shape how

players perceive and respond to different forms of harassment. The perceived severity of gender discrimination and whether players have been victimized appear to have a greater impact on their tendency to ruminate than their gender.

Finding 2: Rumination is more influenced by the perceived severity of discrimination and victimization than by gender. However, female and gender non-conforming participants tend to ruminate more, especially in response to severe verbal harassment.

C. RQ3: Impact of Gender Discrimination and Rumination

Our findings support McLean and Griffiths [5], showing that harassment victims, especially female players, are more likely to leave gaming groups and feel discouraged from playing. However, unlike previous research suggesting that female players are perceived as outsiders [6], [7], [9], [11], [22] – both by others and themselves – our findings show that female players, like participants from other groups, similarly self-identify as gamers.

Overall, rumination increases discouragement from gaming due to gender discrimination, aligning with Fox and Tang [9], who found it contributes to withdrawal, and Cote et al. [11] who identified leaving online gaming as the primary coping strategy of female players. This effect persists despite shifts in gender distribution, and is most pronounced among gender non-conforming players. While gender does not directly influence perceived harassment severity, marginalized players who experience harassment tend to ruminate more, increasing their likelihood of withdrawing from gaming.

The withdrawal from gaming and identity concealment to avoid harassment may reinforce majority dominance, creating a cycle where marginalized groups hide while the majority remains unaware of their presence. This pattern aligns with the hypothesis of maladaptive boundary crossing proposed by Drenten et al. [25], wherein marginalized individuals navigate online spaces in ways that ultimately limit their visibility and inclusion.

While this cycle risks deepening the imbalance in representation and may reinforce and amplify perceptions of inequality [25], our findings show that participation in external gaming communities does not vary with rumination tendency. This aligns with Ma et al. [6] and McLean and Griffiths [5], highlighting the significance of external gaming communities for players, suggesting that these spaces may provide a form of support and engagement that remains unaffected by in-game experiences of harassment.

Finding 3: Rumination increases discouragement from gaming, especially among marginalized groups, reinforcing the majority of the dominant group as players conceal their identities. However, engagement in external gaming communities remains stable, offering inclusive and encouraging environments.

D. RQ4: Players' Usage of In-game Moderation Tools

Unlike Adinolf and Türkay [23], who found ignoring or muting most common, our results align with Kowert and Cook [20]: reporting was used most often, followed by muting. This difference may reflect greater transparency and normalization of reporting systems since their 2018 study.

Players who perceived gender discrimination as more severe used moderation tools more often. Those facing or witnessing discriminatory behavior particularly relied on reporting offending players, the dominant coping strategy. Nevertheless, uncertainty around reporting remains, as 27% of S1 participants and 54% of S2 participants indicated they do not always report discriminatory behavior, suggesting persistent hesitation in a notable subset of players. This suggests, that while systems might have improved over the years, they may still be perceived as opaque, which has previously been identified as a driver of their underuse [23].

Our findings show differences in moderation tool usage between participants with high and low severity perceptions. Notably, reporting on behalf of another targeted player differed significantly in both previously victimized and broader participant groups, whereas reporting personal victimization differed only in the broader group. However, a significant difference between low- and high-severity raters was only observed in the broader group when it came to victimization of the participants themselves. Previously victimized players may therefore respond more uniformly to harassment regardless of their perceived severity of gender discrimination, potentially due to a normalized or habituated response to being targeted. This aligns with our findings on rumination, suggesting that prior victimization and severity perception shape how players respond to gender discrimination, with higher-severity raters more actively reporting or muting offenders.

Moderation tool use also differed by gender, consistent with McLean and Griffiths [5] and Fox and Tang [9], who found that marginalized groups respond differently to discrimination. In our study, female participants reported discriminatory behavior more often than the dominant group. Notably, when looking at male participants only, the high-severity group muted other players more frequently than those in the low-severity group. Similarly, high-severity rating female participants reported discriminatory players more often than their low-severity rating counterparts. This pattern suggests that members of marginalized groups may cope with and respond to discrimination differently than members of the dominant group. The more severe they perceive gender discrimination to be, the more strongly they appear to seek consequences for discriminatory players through reporting.

Finding 4: Perceived severity of gender discrimination predicts higher use of moderation tools overall. Male players show a stronger increase in muting as severity perception rises, while female players show a stronger increase in reporting. This does not mean women mute less generally, but that severity perception drives reporting behavior more strongly in female players than in male.

E. RQ5: Targeted Players Altering Information and Gender Concealment

Our findings indicate that the perceived severity of gender discrimination among players who previously experienced victimization does not affect whether they hide their gender in COMGs. This effect did neither vary by the communication channels through which discrimination occurred, nor by whether players had previously altered in-game profile details, such as username or profile description. We therefore conclude that the individuals' own assessment of discrimination intensity does not appear to shape their choices to reveal or conceal gender. Interestingly, our results suggest that participants with a stronger sense of belonging in gaming were less likely to alter their identity settings to conceal their gender, and those with lower belonging were more likely to change their username specifically to hide their gender, rather than other profile-based methods. Other factors, like personal preferences or availability of tools integrated into the games, may contribute to this behavior, as noted by Frommel and Mandryk [24].

Our findings diverge from those of Frommel and Mandryk [24] regarding the frequency with which players modify their in-game appearance or username. In their study, concealment of gender was one of the least frequently used coping strategies. In contrast, only one-third of previously victimized participants in our study reported never having made any changes to conceal their gender, with nearly half indicating having altered their username at least once. In addition, two-thirds of participants reported that gender-based discrimination had significantly or completely impacted their use of in-game voice chat. Taken together, these patterns suggest that gender concealment may function as a coping mechanism prompted by the anticipation of harassment rather than its severity, with such decisions potentially reflecting an awareness of structural bias rather than responses to specific individual encounters.

Similar to the withdrawal from gaming based on rumination discussed in Section V-C, these trends risk further marginalizing already underrepresented players by limiting their visibility and reinforcing the perception of COMGs as a male-dominated space [25]. Female players may be more willing to present themselves openly when they feel less like a minority. However, as McLean and Griffiths [5] note, the loneliness and anxiety arising from the anticipation of harassment often compel them to maintain anonymity instead, closing the negative feedback loop [25].

Finding 5: Among previously victimized players, perceived severity of gender discrimination does not influence profile modifications for gender concealment, two-thirds have taken action to conceal their gender before regardless. This stability persists across voice and text communications in games. **Severity perception does not drive identity hiding or profile changes to conceal gender.**

F. Threats to Validity

As empirical research, our studies are subject to various threats to validity. Below, we discuss these limitations in alignment with the guidelines of Wohlin et al. [30]:

a) *Construct Validity*: Since this study aims to explore the perception of hateful behavior, our findings might be influenced by social desirability bias. Specifically, some participants might have rated gender discriminatory behavior as severe because they believed it was expected of them. To address this bias, we explicitly asked participants to share their genuine opinions and behaviors. Additionally, we informed them that the study's focus was not on judging their personal character but on understanding the broader behavior of the gaming community. However, we acknowledge that not all participants' responses could be entirely free from bias in this context.

b) *Internal Validity*: Coding procedures like the one in this study are often susceptible to experimenter bias, as different coders may interpret the same data differently. To mitigate this, we involved three researchers in the coding process, one of whom had no previous research experience in gaming and was thus not personally invested in the topic. Additionally, after calculating the inter-rater agreement, we discussed any remaining disagreements until all conflicts were resolved.

c) *Conclusion Validity*: Whenever we analyzed our research data for correlations or statistically significant findings, we employed appropriate statistical methods. Specifically, we utilized Pearson's correlation coefficient, ordinal logistic regression, the chi-square test, Levene's test, Welch's t-test, ANOVA, and the Mann-Whitney U test. All results were interpreted at a 5% significance level. Consequently, we are confident that our findings are well-founded based on the data we collected.

Participants gauged the severity of statements on a 7-point interval scale. We decided to report mean instead of median, to enable a more fine-grained analysis within the large amount of collected statements. When examining participants' answers between those that gave higher than average and those that gave lower than average ratings, we dichotomized the severity rating scale into high- and low-severity groups. This was done to facilitate an interpretable comparison between those that perceive the severity of gender discrimination as high against those who do not. However, doing this might have introduced bias by reducing statistical significance and obscuring non-linear relationships within the data.

We assessed the inter-rater agreement for our coding procedure by calculating the proportional agreement and Fleiss' Kappa. We obtained a proportional agreement of 76.2% and a Fleiss' Kappa value of 80.6%, which suggests substantial to almost perfect agreement between the raters. Importantly, our labeling process permitted multiple labels for each statement, which made achieving a high inter-rater agreement more challenging. Therefore, we are confident that the categories of gender discrimination identified in this study robustly represent the data we collected.

d) *External Validity*: The external validity of our work is constrained by the demographic diversity of our participants. The first survey intentionally included only non-male participants, while the second included all genders, with 61% of

respondents being male. This is consistent with the typical gender distribution in online gaming communities, where male players are the majority. Thus, we believe our findings are representative of larger online gaming communities in terms of gender identity. Geographically, the majority of participants, particularly in the first survey, were from Europe, with a few from the US and nearly none from Asia or Africa. This limits the generalizability of our findings to global gaming communities, as cultural differences may have a substantial impact on perceptions and experiences of online harassment.

Our recruitment method likely introduced selection bias, resulting in an over-representation of players who are aware of or affected by gender discrimination and an under-representation of those who are indifferent to or participate in it. Players that aren't concerned about or even commit gender discrimination themselves could have been turned away from participating in our study due to its topical title. Conversely, we had no access to the players who had already left the communities in which the surveys were distributed. As a consequence, our survey responses might be lacking polar opinions in both directions, which could influence the reported ratings of perceived severity. This imbalance could have skewed the severity ratings, emphasizing the necessity for more representative sampling methods, such as randomized or stratified recruitment, in future research.

VI. CONCLUSION AND FUTURE WORK

Our findings show that players across all gender groups judge gender discrimination severity in largely the same way, with only minor differences related to prior exposure, indicating widespread awareness of the issue. Players' rumination is driven more by perceived severity than by gender identity itself, although marginalized groups tend to ruminate more often, supporting prior research [9]. The data imply that discrimination does not impact all groups in the same way. Players from marginalized groups respond more strongly as the perceived severity increases, most notably through higher rates of reporting other players, while players from dominant groups tend to rely more on muting them. These differences highlight that experiences of and reactions to discrimination are shaped not only by the players' own behavior but also by their social position.

Severe verbal harassment appears central, discouraging players from both gameplay and communication and ultimately contributing to withdrawal. This withdrawal can sustain majority dominance by prompting marginalized players to hide aspects of their identities. The studies indicate that the decision to conceal identity is not shaped by how severe discrimination feels but is a common coping response among players who were previously targeted. The overall patterns show little change over the past eight years, yet engagement in external gaming communities has stayed consistently steady, suggesting that these spaces continue to serve as inclusive and supportive alternatives.

In line with Ma et al. [6] and McLean and Griffiths [5], who emphasize the role of social factors in gaming as both a key motivation for players and a potential positive influence on mental health, we highlight the importance of fostering inclusive and

supportive gaming communities. Creating environments that enable players to openly express and collectively reflect on harassment experiences may lessen rumination and mitigate its harmful effects. Game developers and community leaders can contribute to this by enhancing moderation systems, cultivating inclusive community cultures, and amplifying community-driven initiatives that empower marginalized players and encourage healthier social interactions.

Future research should examine the impact of harassment on different marginalized groups and evaluate how effective community-led interventions are at reducing player rumination. Longitudinal research could further clarify how rumination changes over time, including long-term mental health effects of gender discrimination and alternative explanations, such as personal baseline tendencies for rumination, while extending analysis across a broader range of game genres.

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